March 19, 2026

EXPLORING NEW INSIGHTS ON THE CITY'S URBAN TREE CANOPY AND CLIMATE CHANGE

PHASE TWO – ADDITIONAL ANALYSIS

Revisit the original data. **There is no additional data set** provided for Phase II. The resource links are embedded in this document and will be linked to the competition website. You may pull some additional data and information from these documents if you think it will help.

BACKGROUND

You have already analyzed the City of Hamilton's tree inventory to understand the impacts of climate change on the urban forest. Now, it is time to take your exploration further. The City's Urban Forest Strategy sets a long-term vision for maintaining and enhancing the urban tree canopy, and understanding climate-related risks remains critical. Trees are a valuable city asset that contribute to climate adaptation by absorbing carbon, reducing flood risks, and moderating temperatures. However, the urban forest is also vulnerable to extreme weather, droughts, and long-term environmental changes.

NEW DIRECTIONS FOR EXPLORATION

Instead of repeating previous analyses, focus on identifying new patterns, relationships, and potential strategies. Consider questions that go beyond vulnerability, such as how tree canopy coverage could influence local heat islands, how different neighborhoods might benefit differently from tree growth, or how species diversity can improve overall resilience. Think about linking tree data with other city datasets, such as population density, infrastructure, or air quality, to uncover broader insights.

PROJECT GOALS

Your goal is to explore novel questions using the City's tree inventory and other relevant climate data. This might include assessing the potential benefits of planting new species in specific areas, identifying locations where tree loss would have the largest impact on energy

consumption, stormwater management, or evaluating long-term planning scenarios under different climate projections. Consider how adaptive strategies can be prioritized for maximum impact in the short, medium, and long term.

POSSIBLE AREAS TO EXPLORE

Some ideas to guide your analysis include:

- How can tree planting or maintenance strategies optimize community-level benefits, such as shade, air quality, or stormwater reduction?
- Are there emerging risks from pests, disease, or invasive species that may be amplified by climate change?
- How does tree canopy distribution relate to socioeconomic factors, and what equity considerations should guide future planning?
- Can predictive models help forecast tree health or survival under different climate scenarios?

DATASETS AND CONSIDERATIONS

You will continue to use the City's tree inventory datasets. Remember, the data may require cleaning or transformation before it is ready for analysis, so plan for data preparation as part of your workflow. Be creative and explore angles that can provide new insights into Hamilton's urban forest strategy.

DATA FILES AND LINKS

2026 Data Links – HEAD Competition | Mohawk College

Password: @phase2*HEAD26

PRESENTATION INSTRUCTIONS

Your team presentation will consist of a live 20-minute oral delivery to a panel of judges, followed by a 10-minute Q&A opportunity with the judges. In your presentation, you should tell a story consistent with your analysis that defines your problem, explains your data, methodology, and results, and provides your recommendations or other conclusions.

An electronic copy of your supporting presentation materials is due by 9:00 AM on Friday, March 19, 2026, and can be uploaded to the team folder on the shared drive from Phase I. Judges will have the opportunity to review the presentation materials during your presentation. You may NOT edit your submissions after 9:00 AM. This may lead to disqualification.

You may use any software/tools for your presentation, but you will need to provide some form of PDF output or other easily readable format for judges to review that does not require special software. You do NOT have to use PowerPoint for your presentation. For example, you may use active reports from Power BI, KNIME, and Tableau presentations, if PDF copies of the presentations are made available by 9:00 AM.

Presentations will start at 10:00 AM with 30 minutes set aside for each presentation and a question period. Presentations will be stopped at 20 minutes to allow 10 minutes for questions, so manage your time accordingly.

You may present using your own laptop, or if presenting a PowerPoint presentation, your submission will be pre-loaded on a provided laptop.

Student teams will be able to observe the projects of other teams, but once again, they may not modify their presentation materials once submitted.

Please remember that this is a student competition. Mentors may be used to provide guidance and recommendations, but all analysis and work should be performed by the students.

Best of luck to all the teams!

Sincerely,
2025 HEAD Competition Organizing Committee

- **Please do not share this problem with anyone except your teammates.**
- **Sharing and distributing this problem information is expressly prohibited.**

Criteria	Description
Originality and Creativity	Creativity and originality of logic. Timeliness and uniqueness of ideas. Novel thinking or approach to describe the problem and/or solution.
Organization	Objectives/goals are clearly stated. Methods are appropriate for achieving goals. Results are clearly presented. Thoughts and ideas flow in a logical manner. Results accomplish purpose of the project.
Presentation	Good body posture exhibited. Good eye contact with audience maintained. Good diction; good articulation.
Knowledgeability	Exhibited knowledge of subject matter. Answer questions with confidence.
Media Quality/Neatness	Neat slides. Visual materials are easy to read. Good slide layout with appropriate font selection, sizing, and colors. No obvious typographical errors.
Overall Presentation	Materials are organized with smooth transitions (i.e., Introduction, Methods, Results, Discussion). Style and language are appropriate. Charts, graphs, and/or illustrations are appropriate.
Problem Definition	Problem is clearly defined and communicated.
Methodology/Approach	Clear and logical explanation of method used to prepare and analyze data. Appropriateness of approach.
Model and Analysis	Appropriateness of model and quality of the analysis. Demonstration of applied statistical models.
Conclusion/Recommendations	Clearly defined and clearly communicated results and/or recommendations. Logical results and recommendations based on analysis